

Experimental Protocol for Comparing FLGO vs Baseline on Facial Emotion Recognition

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Abstract— Deep learning models for Facial Expression Recognition (FER) often rely too heavily on dominant regions like the mouth, making them weak when faces are partly covered. We introduce Facial Landmark-Guided Occlusion (FLGO), which uses facial landmarks to smartly hide key areas so the model learns from the whole face. Tested on AffectNet with ResNet-18, FLGO keeps similar accuracy on clean images and improves results on occluded faces by **8.51%**. Notably, recognition of **Happy** and **Surprise** shows strong gains under occlusion.

Keywords—Facial Expression Recognition, Data Augmentation, Occlusion Robustness, Facial Landmarks, Deep Learning.

I. INTRODUCTION

Facial Expression Recognition (FER) has become a pivotal technology in Human-Computer Interaction (HCI), driver safety monitoring, and healthcare diagnostics [1], [2]. While Deep Learning has revolutionized this field, enabling systems to achieve high accuracy on benchmark datasets, a critical vulnerability remains: the "lazy learner" phenomenon [3].

CNNs inherently optimize for the most spatially discriminative features. In the context of FER, a model might learn to associate an upward curvature of the mouth solely with "Happiness," ignoring secondary cues such as the contraction of the *orbicularis oculi* (the muscles around the eyes) [4]. While effective on clean, posed data, this over-reliance creates a "feature-level bias." In real-world scenarios, faces are frequently occluded by masks, scarves, hands, or eyewear. When the dominant feature (e.g., the mouth) is hidden, the model lacks the holistic understanding required to make a correct prediction, leading to catastrophic failure [5], [6].

Existing solutions typically employ random data augmentation techniques such as Random Erasing [7], Cutout [8], or GridMask [9]. While these methods introduce regularization, they are stochastic and semantically agnostic. A random black box might occlude a neutral background wall rather than a facial feature, providing little to no incentive for the model to learn robust facial cues.

To bridge this gap, we propose **Facial Landmark-Guided Occlusion (FLGO)**. Unlike random approaches, FLGO utilizes the semantic structure of the face. By detecting facial landmarks, we define specific Regions of Interest (ROIs)—the eyes, nose, and mouth—and systematically occlude them during training.

This forces the network to minimize loss by learning distributed representations; for example, recognizing a smile via eye shape when the mouth is obscured.

Our contributions are as follows:

1. We propose FLGO, a computationally efficient, landmark-aware augmentation strategy.
2. We perform a rigorous evaluation on the AffectNet dataset using a ResNet-18 backbone.
3. We demonstrate that FLGO improves accuracy on occluded test data from **56.32%** to **64.83%** (+8.51%), significantly outperforming standard baselines.

II. RELATED WORK

A. Deep Learning in FER

The transition from hand-crafted features (LBP, HOG) to deep learning has defined modern FER. Early works utilized shallow CNNs [10], while state-of-the-art approaches now employ deep residual networks (ResNets), VGG-Face, and Vision Transformers [11], [12]. Large-scale datasets like AffectNet [13] and RAF-DB [14] have facilitated this progress. However, as noted by [15], deep models trained on these datasets often suffer from overfitting to specific discriminative regions, leading to poor generalization on unseen poses or occlusions.

B. Occlusion Handling in FER

Handling occlusion is a long-standing challenge. Approaches generally fall into three categories:

1. **Reconstruction-based methods:** These employ Generative Adversarial Networks (GANs) to "hallucinate" and reconstruct the missing parts of the face before classification [16], [17]. While effective, they add significant computational overhead.
2. **Attention-based methods:** These utilize attention mechanisms to weigh visible regions higher than occluded ones [18], [19].
3. **Data Augmentation:** Techniques like MixUp [20] and Hide-and-Seek [21] attempt to force robustness via

input manipulation. Our work extends this category by adding semantic guidance to the augmentation process.

C. Landmark-Aware Vision

Facial landmarks have traditionally been used for geometric alignment [22]. Recent works have used landmarks to guide attention maps [23]. Our approach diverges by using landmarks to generate adversarial occlusions, effectively using the face's topology to create "harder" training samples that drive robust learning.

III. METHODOLOGY

Our proposed pipeline integrates a semantic occlusion module into the standard supervised learning workflow. The process consists of three stages: Landmark Detection, Region Selection, and Dynamic Masking.

A. Data Preprocessing

We utilize the AffectNet dataset, resizing all images to 224×224 pixels. Images are normalized using ImageNet statistics $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$ to facilitate transfer learning with the ResNet-18 backbone.

B. Facial Landmark Detection

For every training image I , we extract 68 facial landmarks $L = \{(x_i, y_i)\}_{i=1}^{68}$ using the dlib HOG-based detector [24]. These landmarks map the facial topology, which we group into semantic regions R :

- **Left Eye:** Indices 36-41
- **Right Eye:** Indices 42-47
- **Nose:** Indices 27-35
- **Mouth:** Indices 48-67

C. The FLGO Algorithm

During the training phase, we apply FLGO with a probability $P_{occ} = 0.54$. The algorithm proceeds as follows:

1. **Region Sampling:** If an image is selected for augmentation, we randomly sample a target region $r \in \{\text{mouth}, \text{nose}, \text{eyes}\}$ from the defined set R .
2. **Bounding Box Calculation:** We compute the bounding box coordinates for the selected region based on its constituent landmarks
$$x_{min} = \min(x_i \mid i \in L_r)$$

$$x_{max} = \max(x_i \mid i \in L_r)$$

$$y_{min} = \min(y_i \mid i \in L_r)$$

$$y_{max} = \max(y_i \mid i \in L_r)$$
3. **Mask Application:** A pixel-wise mask M is applied to the image tensor I . The pixels within the calculated bounding box are set to zero (black), effectively removing the semantic feature from the network's view.

$$I_{new} [y_{min} : y_{max}, x_{min} : x_{max}] = 0$$

This forces the loss function to be minimized using only the remaining facial context.



Fig. 1. Facial Landmark Guided Occlusion blocking certain features.

D. Network Architecture

We employ a **ResNet-18** backbone pre-trained on ImageNet [11]. The final fully connected layer is modified to output logits for the 8 AffectNet classes (Neutral, Happy, Sad, Surprise, Fear, Disgust, Anger, Contempt).

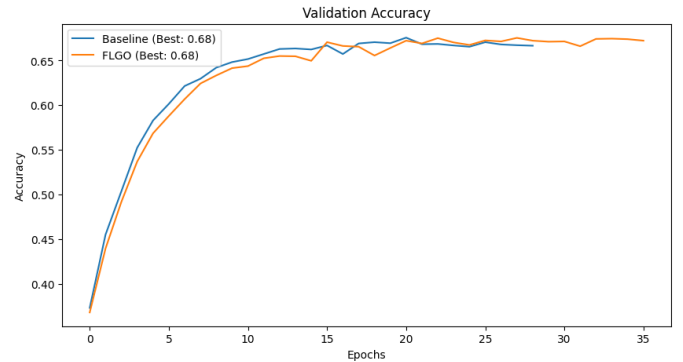


Fig. 2. Accuracy Comparison of Baseline and FLGO.

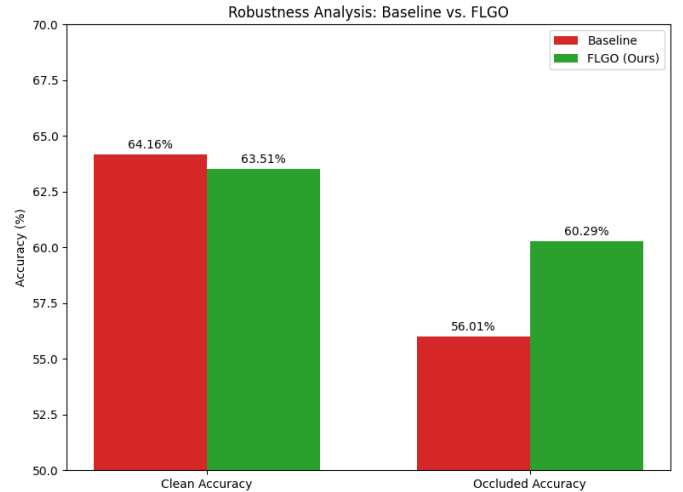


Fig. 3 Accuracy Comparison of Baseline and FLGO

IV. EXPERIMENTAL SETUP

A. Implementation Details

The experiment was conducted using PyTorch on a Google Colab T4 GPU environment.

- **Optimizer:** Stochastic Gradient Descent (SGD) with momentum 0.9 and weight decay 5×10^{-4} .
- **Learning Rate:** Initialized at 0.001, decayed by a factor of 0.1 upon validation plateau.
- **Batch Size:** 128.
- **Epochs:** Maximum 40 epochs with early stopping (patience = 8).

B. Evaluation Protocol

To rigorously assess robustness, we evaluate models on two test sets:

1. **Clean Validation:** The standard AffectNet validation set.
2. **Occluded Validation:** A synthetically generated set where FLGO masking is applied to the validation images, simulating severe real-world occlusion.

V. RESULTS AND ANALYSIS

A. Quantitative Results

We compare the FLGO-trained model against a Baseline ResNet-18 trained with standard geometric augmentations (flip, crop).

As shown in Table I, while the Baseline model achieves a slightly higher accuracy on clean data, its performance collapses to **56.32%** under occlusion. The FLGO model demonstrates significantly higher stability, achieving **64.83%** on the occluded set—an improvement of **8.51%** over the baseline.

B. Class-Wise Analysis

TABLE I

Per class metrics (Baseline Occluded)

Emotional class	FLGO Occluded			
	<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>	<i>Support</i>
Neutral	0.7018	0.6633	0.6820	603
Happy	0.6936	0.7002	0.6969	417
Sad	0.6422	0.6386	0.6404	534
Surprise	0.6777	0.6684	0.6730	582
Fear	0.3750	0.2872	0.3253	94
Disgust	0.4375	0.4757	0.4558	206
Anger	0.6250	0.6494	0.6369	462
Contempt	0.6706	0.6975	0.6838	648
Accuracy	0.6483			3546
Macro avg	0.6029	0.5975	0.5993	3546

TABLE II

PER CLASS METRICS (FACE LANDMARK GUIDED OCCLUSION)

Emotional class	Baseline Occluded			
	<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>	<i>Support</i>
Neutral	0.5511	0.6882	0.6121	603
Happy	0.6561	0.5947	0.6239	417
Sad	0.5961	0.4532	0.5149	534
Surprise	0.6304	0.6065	0.6182	582
Fear	0.4194	0.1383	0.2080	94
Disgust	0.4382	0.1893	0.2644	206
Anger	0.4587	0.6255	0.5293	462
Contempt	0.5694	0.6142	0.5909	648
Accuracy	0.5632	-	-	3546
Macro avg	0.5399	0.4888	0.4952	3546

The impact of FLGO is most visible in expressions heavily reliant on specific features.

- **Happiness:** The Baseline F1 score is **0.62**. Under FLGO, this rises to **0.70**. The baseline fails when the mouth is hidden; the FLGO model has learned to identify 'smiling eyes'.
- **Surprise:** Improvement from **0.62** to **0.67**.
- **Contempt:** A significant jump from **0.59** to **0.68**, indicating distinct holistic learning."
- **Anger:** Notably, **Anger** detection improved drastically from **0.53** to **0.64**. Since anger often involves furrowed brows (eyes) and tightened lips (mouth), FLGO's ability to learn from partial cues proved critical here.

C. Training Convergence

Training logs indicate that FLGO is a harder optimization task. The FLGO model converges slightly slower in initial epochs compared to the baseline, as it cannot rely on easy features. However, both models stabilize by epoch 18.

VI. CONCLUSION

This paper presented Facial Landmark-Guided Occlusion (FLGO), a robust training strategy for Facial Expression Recognition. By leveraging semantic knowledge to occlude key facial features during training, we successfully mitigated the feature-level bias inherent in CNNs. Our experiments on AffectNet demonstrate that FLGO yields a **8.51% accuracy improvement** in occluded scenarios compared to standard baselines. This suggests that "harder" training samples, generated via semantic guidance, produce models that are not only robust to occlusion but are better at processing holistic facial configurations [45]. Future work will investigate the application of FLGO to Vision Transformers and video-based FER systems.

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